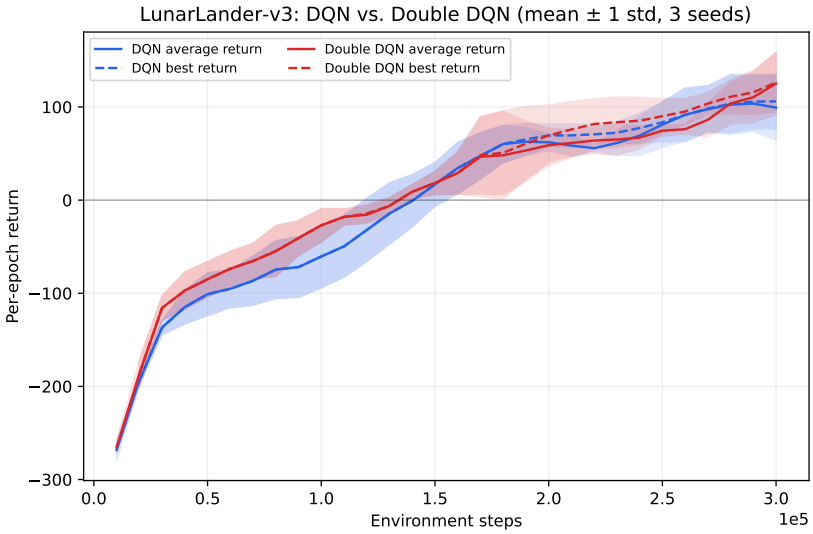
Assignment 3: Q-Learning and Actor-Critic Algorithms

Andrew ID: `jingshup`
Collaborators: None.
NOTE: Please do **NOT** change the sizes of the answer blocks or plots.

1 Question 1: DQN and DDQN (LunarLander-v3) - [7.5 points]

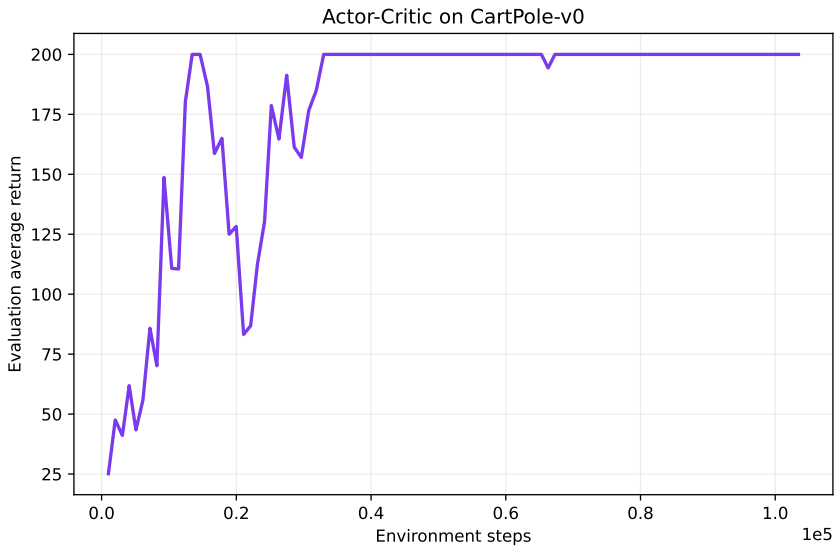
Question 1



Mean performance across seeds 1–3; shading shows one standard deviation. Both methods use $\gamma = 0.99$, target updates every 2,000 steps, and epsilon decay from 1.0 to 0.02 over the first 90,000 steps.

2 Question 2: Actor-Critic (CartPole-v0) - [3.75 points]

Question 2



Actor-critic reaches the CartPole-v0 maximum return of 200.

3 Question 3: Actor-Critic (InvertedPendulum-v4) - [3.75 points]

Question 3



Evaluation return for actor-critic with 10 target updates and 10 gradient steps per target.